LAND PARCEL AREA AND PERIMETER CALCULATOR
PROJECT REPORT

Programming Language: Python

Development Environment: Visual Studio Code (VS Code)

Project Type: Geomatics / Land Surveying Application

NAMES

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1. INTRODUCTION

Land surveying involves collecting and processing information about the location, size and shape of land parcels. One of the common tasks in surveying is determining the area and perimeter of a land parcel from the coordinates of its boundary points.

Traditionally, area and perimeter calculations can be performed manually using mathematical formulas. However, when a parcel contains many coordinate points, manual calculations can become time-consuming and may result in arithmetic errors.

To address this problem, our group developed a Land Parcel Area and Perimeter Calculator using Python. The program allows a user to enter the coordinates of the boundary points of a land parcel. The program then processes the coordinates and automatically calculates the area and perimeter of the parcel.

The program uses the Shoelace Formula to calculate the area and the distance formula to calculate the perimeter. The calculated results are displayed to the user and can also be saved for future reference. The project demonstrates how programming can be applied to solve a practical problem in land surveying and geomatics.

2. PROBLEM STATEMENT

Land surveyors and students may need to calculate the area and perimeter of land parcels from coordinate data. When these calculations are performed manually, especially for

parcels with many boundary points, the process can take considerable time and may lead to calculation errors.

There is therefore a need for a simple computer-based application that can accept land parcel coordinates, process them automatically, and produce meaningful area and perimeter results.

Our project addresses this problem by developing a Python application that automatically calculates the area and perimeter of a land parcel when its boundary coordinates are provided.

3. PROPOSED SOLUTION

The proposed solution is a Python-based Land Parcel Area and Perimeter Calculator. The application accepts the coordinates of the boundary points of a land parcel as input and processes them using appropriate mathematical formulas.

The program performs the following tasks:
1. Accepts the coordinates of land boundary points.
2. Stores the coordinates for processing.
3. Calculates the distance between consecutive boundary points.
4. Adds the distances to obtain the perimeter.
5. Applies the Shoelace Formula to calculate the area.
6. Converts the area into other useful units where required.
7. Displays the calculated results.
8. Saves the results in a text file.

The proposed solution reduces manual calculation, saves time, and makes the calculation process easier to perform.

4. OBJECTIVES

4.1 General Objective

To develop a Python-based application that automatically calculates the area and perimeter of a land parcel from its boundary coordinates.

4.2 Specific Objectives

1. To develop a Python program that accepts coordinates of land parcel boundary points.
2. To calculate the distance between consecutive coordinate points.
3. To automatically determine the perimeter of the land parcel.
4. To calculate the area of the land parcel using the Shoelace Formula.
5. To provide meaningful and understandable results to the user.
6. To save the calculated results for future reference.

7. To test the program using sample coordinate data to determine whether it works correctly.

5. METHODOLOGY

The project was developed using Python in the Visual Studio Code (VS Code) development environment. The development process involved problem identification, algorithm design, coding, testing and evaluation.

5.1 Input of Coordinates

The user provides the coordinates of the boundary points. Each point contains an X-coordinate and a Y-coordinate. In a surveying context, these can represent Easting and Northing coordinates.

Example:
(100, 100)
(200, 100)
(200, 200)
(100, 200)

5.2 Area Calculation

The program uses the Shoelace Formula to calculate the area of the polygon formed by the coordinates:

$$A = 1/2 \, |\Sigma(x_i y_{i+1} - y_i x_{i+1})|$$

The coordinates are processed in their boundary order. The absolute value makes the final area positive regardless of whether the coordinates were entered clockwise or anticlockwise.

5.3 Perimeter Calculation

The program calculates the distance between each consecutive pair of coordinates using the distance formula:

$$d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

The distances of all boundary sides are added together. The program also connects the last coordinate back to the first coordinate so that the complete parcel boundary is included.

5.4 Processing and Output

After the coordinates are entered, the program processes them and displays the calculated area and perimeter. The results can also be saved in a text file for future reference.

6. PROGRAM DEVELOPMENT

The application was developed in Python because Python is relatively simple to learn and provides the programming and mathematical features required for the project.

The major programming concepts used include variables, lists, user input, loops, mathematical operations, functions and file handling.

The coordinates entered by the user are stored in a list. Loops are then used to move through the coordinate points and perform the required calculations.

For perimeter calculation, the program uses the distance between consecutive points. For area calculation, it applies the Shoelace Formula. The results are then displayed and saved in a text file.

The program was tested using a square parcel with the coordinates (100,100), (200,100), (200,200) and (100,200).

7. TESTING AND RESULTS

The developed application was tested using sample coordinate data to determine whether it was functioning correctly.

Test Case 1: Square Parcel

Coordinates:
P1 = (100,100)
P2 = (200,100)
P3 = (200,200)
P4 = (100,200)

Expected Area:
$100 \times 100 = 10{,}000\ m^2$

Expected Perimeter:
$100 + 100 + 100 + 100 = 400\ m$

Program Result:
Area = $10{,}000\ m^2$
Perimeter = 400 m

Status: PASS

The test demonstrated that the program correctly calculated both the area and perimeter for the sample parcel. Additional coordinate sets can be used to verify the application with different land parcel shapes.

8. CHALLENGES ENCOUNTERED

During development, the group encountered several challenges.

1. Understanding the mathematical formulas. The group had to understand how the Shoelace Formula and distance formula could be converted into Python instructions.

2. Entering coordinates correctly. Incorrect or incomplete coordinate input can affect the final results. The program was therefore tested using different coordinate sets.

3. Closing the parcel boundary. The last point must be connected back to the first point when calculating perimeter. The program was designed to include this connection.

4. Testing the program. The group needed to establish whether the results were correct. Known examples, such as a square with sides of 100 m, were used to compare manual calculations with program results.

9. IMPORTANCE OF THE PROJECT

The project demonstrates how programming can be applied to practical problems in surveying and geomatics. The application can reduce manual calculations, save time, reduce arithmetic errors, make coordinate-based calculations easier, and help students understand the relationship between programming and surveying.

Although the application was developed as an academic project, the same principle can be extended to more advanced surveying and GIS applications.

10. CONCLUSION

The project successfully developed a Python-based Land Parcel Area and Perimeter Calculator. The application accepts the coordinates of land parcel boundary points and processes them to calculate the area and perimeter automatically.

The area is calculated using the Shoelace Formula, while the perimeter is calculated using the distance formula. Testing using sample coordinates showed that the program can produce the expected results.

The project demonstrates that Python programming can be used to automate basic land surveying calculations. It also provided the group with practical experience in Python programming, coordinate processing, mathematical calculations, testing and application development.

11. FUTURE IMPROVEMENTS

The application can be improved in the future by adding:
1. A graphical user interface (GUI).

2. A map showing the entered land parcel.
3. Automatic detection of incorrect coordinates.
4. Importing coordinates directly from CSV or Excel files.
5. Exporting results to PDF.
6. Calculation of additional surveying measurements.
7. Integration with GIS software.
8. Support for larger and more complex land parcels.
9. Better error handling for invalid user input.
10. Online deployment so that users can access the calculator through a web browser.